

UnifAI Complete Platform Master Technical Manual

Exhaustive Technical & Operational Specification Covering All 40 Features Across 9 Pillars (Thanglish + English)

1. Master System Architecture Overview

This master document represents the single source of truth for the entire UnifAI platform, covering all 40 features across 9 core pillars: Observability, Models & Traffic Control, MCP Gateway & Tool Execution, Prompt Management & Playground, Plugins & Extensibility, Governance & Identity, Security Guardrails, High Availability Infrastructure, and Adaptive Routing. Every feature is detailed from both an intuitive non-technical perspective (real-world analogies, plain Thanglish explanation, business value) and an in-depth technical engineering perspective (FastHTTP architecture, endpoints, UI layouts, and connection structure maps).

2. Comprehensive Feature Breakdown (40 Features Across 9 Pillars)

Pillar: Observability & Telemetry (Observability (Kangaanippu & Seyalthiran Paguppaayvu))

Company-la nadakkura ellam AI transactions, cost, latency, token spend, matrum error logs-ah real-time-la track panni external monitoring tools (Datadog, Kafka) kooda connect pandra control plane.

Dashboard — Dashboard (Main Metrics & Analytics Cockpit)

Route: `/workspace/dashboard`

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Vimanathoda Cockpit allathu luxury car speedometer dashboard mathiri.
- **Vilakkam:** Car ottum pothu speed, petrol level, engine condition ore meter-la theriyura mathiri, company-oda total AI spend, ethana lakh requests vandhathu, entha AI model fast-ah irukku, entha team athigama spend pandranga nu ore glance-la real-time graphs vazhiya pakkalam.
- **Business Value:** Immediate budget transparency, cost spike prevention, and executive performance reporting.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** WebSocket connection (`useWebSocket``) synchronized with TanStack Query. FastHTTP lock-free atomic counters (`sync/atomic`) aggregate time-series buckets into PostgreSQL. Computes p50, p90, p95, p99 latencies and Time-To-First-Token (TTFT).
- **Endpoints:** GET `/api/v1/dashboard/stats`, GET `/api/v1/dashboard/histogram`, WS `/api/v1/ws/dashboard`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Time Range Selector (1h, 24h, 7d, 30d, Custom)
- Timezone Toggle (UTC / Local)
- Export Popover (CSV/JSON/PNG)
- Filter Sidebar Toggle

2. Mathiya Koorugal & Tables (Tabs & Views):

- Overview Tab (Total Spend, Requests, Tokens, p95 Latency)
- Provider Usage Tab
- Model Rankings Tab
- Dimension Rankings (x-uf-dim-*)
- MCP Tool Metrics

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- Line vs Bar Chart Toggle
- Model Performance Leaderboard
- Live WebSocket Status Indicator

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Aggregates raw telemetry from LLM Logs and FastHTTP Proxy.
- **Iyakkum Koorugal:** Alerts finance managers and provides data for Budget enforcement.
- **Production Use Case:** CFO weekly review analyzing whether the company is staying within the \$20,000 monthly AI budget.

LLM Logs — LLM Logs (AI Request & Response Transaction Ledger)

Route: </workspace/logs>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Bank passbook statement allathu CCTV footage log book mathiri.
- **Vilakkam:** Customers and employees AI kitta enna kelvi kettanga, AI enna bathil sonnathu, ethana tokens aachu, evlo dollar bill aachu nu second-by-second capture seiya complete transaction ledger.
- **Business Value:** 100% auditability, prompt debugging, compliance evidence, and customer dispute resolution.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** High-speed batch-insert pipeline into PostgreSQL `llm\_logs` table. Stores prompt text, model response, TTFT, token usage, status codes, and virtual key tags with sub-millisecond overhead.
- **Endpoints:** GET `/api/v1/logs`, GET `/api/v1/logs/{id}`, POST `/api/v1/logs/export`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Search Input (Prompt/Response query)
- Column Visibility Picker
- Export Button (CSV/JSON)
- Auto-Refresh Dropdown

2. Mathiya Koorugal & Tables (Tabs & Views):

- Logs Table: Status Code, Model, Latency ms, Total Tokens, Spend \$, Virtual Key, User ID, Timestamp

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- Log Detail Sheet: Formatted Message Tree, Raw Headers, Guardrail Verdicts, Raw JSON Copy Button

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Intercepts all calls passing through FastHTTP Proxy Gateway.
- **Iyakkum Koorugal:** Feeds live data into Dashboard, Connectors, and Budgets & Limits.

■ **Production Use Case:** Customer support team debugging why an AI agent gave an incorrect shipping estimate to a VIP customer.

MCP Logs — MCP Logs (Autonomous Agent Tool Call Audit)

Route: </workspace/mcp-logs>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Factory supervisor duty register mathiri.
- **Vilakkam:** AI Agent thana mudiveduthu company database-ah paathucha, email anupucha, file-ah delete pannucha nu AI seitha ovoru external action-aiyum monitor seiya tool execution audit book.
- **Business Value:** Autonomous agent accountability, security verification, and third-party API call monitoring.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Model Context Protocol (MCP) JSON-RPC 2.0 frame capture engine. Records tool names, arguments, execution duration, and stdout/stderr outputs across stdio and SSE transports.
- **Endpoints:** GET `/api/v1/mcp/logs`, GET `/api/v1/mcp/logs/{id}`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Filter by MCP Server
- Tool Name Search
- Execution Status Filter (Success/Fail/Timeout)

2. Mathiya Koorugal & Tables (Tabs & Views):

- MCP Logs Table: Server Name, Tool Executed, Duration ms, Payload Size KB, Invocation Status

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- Tool Execution Drawer: Input JSON Arguments, Tool Response JSON, Error Stack Trace

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Intercepts tool calls executed in MCP Gateway.
- **Iyakkum Koorugal:** Feeds tool reliability statistics into MCP Gateway and Dashboard.

■ **Production Use Case:** Verifying that an automated sales agent only searched the CRM database and did not modify any customer records.

Browser AI — Browser AI (Employee Web ChatGPT DLP Proxy)

Route: </workspace/browser-ai>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Office entrance security guard & bag scanner mathiri.
- **Vilakkam:** Employees company laptop-la web ChatGPT allathu Claude use pannum pothu, company passwords, client details, allathu confidential code-ah copy-paste pannu leak seiyatha mathiri thadukkura security wall.
- **Business Value:** Prevents corporate data leaks to public web AI services without having to ban AI tools for employees.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Client-side mitmproxy daemon intercepting web traffic to `chatgpt.com` and `claude.ai`. Inspects clipboard paste events and file uploads using local regex and Presidio DLP algorithms.
- **Endpoints:** GET `/api/v1/browser-ai/sessions`, GET `/api/v1/browser-ai/violations`, POST `/api/v1/browser-ai/sync`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Proxy Status Badge (Active/Inactive)
- Policy Switcher (Block / Redact / Audit)
- Extension Heartbeat

2. Mathiya Koorugal & Tables (Tabs & Views):

- Intercepted Sessions Tab
- DLP Violations Tab
- Attachment Scans Tab

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- Violation Incident Viewer: Diff showing original confidential text vs redacted asterisk text

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Employee browser extensions and local proxy daemons.
- **Iyakkum Koorugal:** Blocks confidential pastes and logs security violations to Audit Logs.

■ **Production Use Case:** Employee accidentally copying a customer's credit card number into ChatGPT gets warned and the data is masked automatically.

Connectors — Connectors (External Observability & Telemetry Pipelines)

Route: </workspace/observability>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Water pipeline network mathiri (Thanniya thevaiyana tank-ku anuppura mathiri).
- **Vilakkam:** UnifAI-la create aagura AI logs and metrics-ah company-oda main monitoring tools (Datadog, Splunk, Dynatrace, Apache Kafka, Google BigQuery)-ku automatic-ah push seiyura export pipelines.
- **Business Value:** Zero data silos; seamless integration with enterprise security and billing infrastructure.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Async streaming telemetry push engine. Batches events into configurable queue workers with retry backoff and Dead Letter Queue (DLQ). Pushes via OpenTelemetry (OTel), HTTP webhooks, or Kafka producers.
- **Endpoints:** GET `/api/v1/observability/connectors`, POST `/api/v1/observability/connectors`, DELETE `/api/v1/observability/connectors/{id}`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Add Connector Button
- Connector Type Filter (Datadog, Kafka, BigQuery, OTel)
- Health Indicator

2. Mathiya Koorugal & Tables (Tabs & Views):

- Connectors List Cards with endpoint URL, flush interval ms, and delivery status badge

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- AddConnectorDialog: Endpoint URL, Bearer Token, Batch Size (100-5000), Send Test Ping Button

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Subscribes to FastHTTP Proxy event stream and LLM Logs.
- **Iyakkum Koorugal:** Streams enterprise telemetry to corporate BigQuery and Kafka topics.
- **Production Use Case:** Streaming every AI transaction into corporate BigQuery for monthly automated billing invoices.

Logs Settings — Logs Settings (Log Governance & Retention Policies)

Route: </workspace/config/logging>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Office record room archival & paper shredding rules mathiri.
- **Vilakkam:** Pazhaya AI logs-ah ethana naalaikku apram automatic-ah delete pannanum (e.g. 30 days retention), yaarachum PII data anupuna epdi mask pannanum, cold storage S3-ku epdi move pannanum nu set pandra control room.
- **Business Value:** 100% GDPR, HIPAA, and SOC-2 data compliance; saves huge database storage costs.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Log lifecycle engine. Schedules automated PostgreSQL partition drops, triggers `VACUUM` maintenance, manages multi-part uploads to AWS S3 Glacier, and controls traffic sampling rates (1% to 100%).
- **Endpoints:** GET </api/v1/config/logging>, PUT </api/v1/config/logging>, POST </api/v1/config/logging/purge>

■■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Master Logging Switch
- Data Retention Input (Days)
- Traffic Sampling Slider (0-100%)

2. Mathiya Koorugal & Tables (Tabs & Views):

- Privacy Controls: Mask PII Switch, Store Prompts Toggle, S3 Cold Storage Backup Bucket input

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- Purge Expired Logs Now Button
- Save Logging Settings Button

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Administrative configuration inputs.
- **Iyakkum Koorugal:** Enforces storage cleanup in PostgreSQL and automated masking in LLM Logs.

■ **Production Use Case:** Setting a strict 30-day retention rule so user chat history is permanently wiped to comply with European privacy laws.

Pillar: Models & Traffic Control (Models (AI Model Catalog, Routing & Resilience))

Company-oda complete AI models inventory, vendor API keys, smart routing rules, cost triage, and zero-downtime failover systems.

Model Catalog — Model Catalog (AI Models Inventory & Menu)

Route: </workspace/model-catalog>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Amazon product catalog allathu hotel menu card mathiri.
- **Vilakkam:** Company-la configure pannu irukkura ellam AI models (GPT-4o, Claude 3.5 Sonnet, Gemini 1.5, Llama 3) list, athoda context length, 24-hour traffic, total cost ellame ore idathula pakkalam.
- **Business Value:** Eliminates duplicate model usage; standardizes AI models across teams; provides full pricing visibility.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** FastHTTP model registry endpoint (`/api/v1/models/catalog`). TanStack Query client caching with real-time token metrics aggregation. Ingests 24h usage telemetry from PostgreSQL.
- **Endpoints:** GET `/api/v1/models/catalog`, GET `/api/v1/models/attributes`, PATCH `/api/v1/models/{id}/attributes`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- 4 Summary Cards (Total Providers, Total Models, Requests 24h, Cost 24h)
- Provider Filter Dropdown
- Overview vs Attributes Tabs

2. Mathiya Koorugal & Tables (Tabs & Views):

- Overview Table: Provider, Models Used, Traffic 24h, Cost 24h
- Attributes Table: Model ID, Display Name, Pricing, Context Window, Output Tokens

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- AttributeSheet: Edit display name, description, context length, modality tags

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Model Providers and LLM Logs.
- **Iyakkum Koorugal:** Supplies model options to Routing Rules and Complexity Router.

■ **Production Use Case:** Auditing and deprecating older, expensive models like GPT-4 32k in favor of cheaper GPT-4o-mini.

Model Providers — Model Providers (AI Vendor Credentials & Gateway Settings)

Route: </workspace/providers>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Telecom SIM card & network switchboard mathiri (Airtel, Jio, Vodafone connections).
- **Vilakkam:** OpenAI, Anthropic, AWS Bedrock, Google Vertex AI, Azure OpenAI, Groq, Ollama aagiya companies-oda connect aaga thevaiyana API keys, secret credentials, network speed, timeout settings configure pandra control panel.
- **Business Value:** Zero vendor lock-in; secure central key vault; support for private on-premise AI models.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Multi-vendor credential management and FastHTTP connection pool. Configures custom base URLs, proxy tunnels, exponential backoff retries, and raw payload flags. Instant connection ping testing.
- **Endpoints:** GET `/api/v1/providers`, POST `/api/v1/providers`, PATCH `/api/v1/providers/{name}`, DELETE `/api/v1/providers/{name}`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Provider Sidebar with status badges (Active/Warning/Error)
- Add Provider Dropdown (+ Add Custom Provider)

2. Mathiya Koorugal & Tables (Tabs & Views):

- Provider Header: Icon, Name, Edit Provider Config button, Delete button
- ModelProviderKeysTableView: Key Name, Masked Key Secret, Quota Limit, Ping Test button

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- ProviderConfigSheet: Concurrency limits, buffer size, network timeouts, retry delays, proxy settings
- AddCustomProviderSheet: Custom base URL and Keyless mode toggle

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Admin inputs.
- **Iyakkum Koorugal:** Powers outbound AI traffic in FastHTTP Proxy and feeds Routing Rules.

■ **Production Use Case:** Safely rotating production OpenAI keys and adding a local private Ollama instance for sensitive data.

Budgets & Limits — Budgets & Limits (Cost Caps & Token Rate Limits)

Route: </workspace/model-limits>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Credit card daily & monthly spend limit mathiri.
- **Vilakkam:** AI use panna panna bill laatchakanakula pogama irukka, ovvoru team-kum, user-kum, athavathu ovvoru AI model-kum monthly budget (\$500) allathu daily token limit set pandra edham. Limit thanduna automatic-ah request stop aagidum.
- **Business Value:** Eliminates bill shock; guarantees 100% budget compliance across departments.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** High-throughput token bucket rate limiter and budget enforcer middleware. Atomic Redis/PostgreSQL counters (`INCRBY`). Pre-flight evaluation rejects requests with HTTP 429 or routes to a cheaper fallback.
- **Endpoints:** GET `/api/v1/governance/model-limits`, POST `/api/v1/governance/model-limits`, DELETE `/api/v1/governance/model-limits/{id}`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Create Model Limit Button
- Search Bar
- Provider Filter
- Scope Filter (Global, Virtual Key, User, Team)

2. Mathiya Koorugal & Tables (Tabs & Views):

- Model Limits Table: Scope Badge, Target Entity, Provider, Model, Budget Progress Bar \$, Token Progress Bar, Reset Duration

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- ModelLimitSheet: Scope Selector, Target pickers, Max Budget \$, Max Tokens, Reset Duration, Threshold Alert Rules (80% warning, 100% hard block)

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Rates from Pricing Overrides and live spend from LLM Logs.
- **Iyakkum Koorugal:** Blocks or reroutes calls in FastHTTP Proxy when quota is exhausted.

■ **Production Use Case:** Giving intern developers a strict \$50/month cap while granting the production app an elastic \$5,000 budget.

Routing Rules — Routing Rules (Intelligent Traffic Steering & Fallbacks)

Route: </workspace/routing-rules>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Railway track switcher mathiri (Train-ah thevaikku etha track-la thiruppura mathiri).
- **Vilakkam:** Oru customer request varum pothu, athu entha AI model-ku poganum nu rules poduvom. Example: Premium user-ku GPT-4o, Free user-ku Claude 3.5 Haiku nu condition base panni smart-ah route pandrathu.
- **Business Value:** Custom user experience; cost optimization based on customer value; zero downtime via fallback chains.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** CEL (Common Expression Language) evaluation engine written in Go. Evaluates conditions against request headers (`x-uf-*`), prompt metadata, user roles, and token counts to dispatch to prioritized target pools.
- **Endpoints:** GET `/api/v1/routing-rules`, POST `/api/v1/routing-rules`, PATCH `/api/v1/routing-rules/{id}`, DELETE `/api/v1/routing-rules/{id}`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Create Routing Rule Button
- Search Input
- Priority Sort

2. Mathiya Koorugal & Tables (Tabs & Views):

- Routing Rules Table: Rule Name, Priority Badge, CEL Condition, Target Model/Provider, Enabled Switch, Actions
- Visual Rule Tree View

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- RoutingRuleSheet: Rule Name, Priority slider (1-100), Visual/Raw CEL Condition Builder, Target Provider/Model, Fallback Priority list

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** FastHTTP Proxy requests and Complexity Router tiers.
- **Iyakkum Koorugal:** Directs traffic to upstream Model Providers and triggers Circuit Breaker on failure.
- **Production Use Case:** Routing all European requests to EU-hosted Azure OpenAI instances for strict GDPR data residency.

Complexity Router — Complexity Router (AI Query Triage & Cost Optimizer)

Route: </workspace/complexity-router>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Hospital Triage System mathiri (Fever-ku junior doctor, surgery-ku specialist mathiri).
- **Vilakkam:** Simple-ana kelvi ('Hi', 'What is Python?') ketta costly GPT-4o thevailla, athukku fast & cheap-ana GPT-4o-mini pothum. Aana periya complex coding allathu math problem ketta mattum o1 allathu Claude Opus-ku automatic-ah anupum. Ithanaala 70% AI bill micham aagum!
- **Business Value:** Saves up to 70% in monthly AI costs with zero compromise on quality.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Multi-factor prompt complexity analysis engine. Computes a continuous complexity score (0.0 to 1.0) using token length heuristics, vocabulary entropy, code regexes (`def`, `class`, `function`), and keyword weighting. Maps into 4 tiers: `SIMPLE`, `MEDIUM`, `COMPLEX`, `REASONING`.
- **Endpoints:** GET `/api/v1/governance/complexity-analyzer`, PUT `/api/v1/governance/complexity-analyzer`, POST `/api/v1/governance/complexity-analyzer/reset`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Save Configuration Button
- Reset to Defaults Button (RotateCcw)
- Docs Link

2. Mathiya Koorugal & Tables (Tabs & Views):

- Progressive okLCH Palette Bar (Simple, Medium, Complex, Reasoning)
- Tier Boundary Inputs (Simple->Med, Med->Comp, Comp->Reason)
- Keyword Lists (TagInput for Code, Reasoning, Simple)

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- Tier Target Model Mappings (Assign default model per tier)
- Interactive Prompt Test Sandbox

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Intercepts incoming requests in FastHTTP Proxy before Routing Rules.
- **Iyakkum Koorugal:** Selects appropriate model tier and logs classification in LLM Logs.

■ **Production Use Case:** Customer support bot where 85% routine queries go to lightweight models and 15% complex technical queries go to reasoning models.

Circuit Breaker — Circuit Breaker (Outage Protection & Auto-Failover)

Route: </workspace/circuit-breaker>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Veetla irukkura Electric MCB / Fuse mathiri.
- **Vilakkam:** OpenAI allathu Anthropic server crash aanaallathu rate limit aagi error vantha, user-ku error screen kaatama, automatic-ah fraction of second-la backup model-ku (e.g. AWS Bedrock allathu Azure)-ku switch pannu 24x7 non-stop-ah vela seiya vaikkum.
- **Business Value:** 99.99% Enterprise High Availability; zero customer-facing downtime during AI vendor outages.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Distributed Circuit Breaker finite state machine (Closed -> Open -> Half-Open). Monitored via real-time polling every 8 seconds (pollingInterval: 8000ms). Detects response signals (HTTP 429/503, 'x-ratelimit-remaining: 0'). Instantly redirects calls to fallback provider with automated cooldown probing.
- **Endpoints:** GET `/api/v1/circuit-breaker/policies`, POST `/api/v1/circuit-breaker/policies`, PUT `/api/v1/circuit-breaker/policies/{name}`, GET `/api/v1/circuit-breaker/state`, POST `/api/v1/circuit-breaker/policies/{name}/reset`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Header with Shield Icon
- Create Policy Button

2. Mathiya Koorugal & Tables (Tabs & Views):

- Circuit Breaker Table: Policy Name, Primary Provider/Model, Fallback Provider/Model, Trigger Condition, Cooldown, State Badge (Closed/Open/Half-Open), Actions

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- Circuit Policy Dialog: Policy Name, Primary/Fallback comboboxes, Response Header Signal matchers, Cooldown duration
- Manual Circuit Reset Button (RotateCcw)

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Monitors HTTP response headers and errors from FastHTTP Proxy.
- **Iyakkum Koorugal:** Reroutes failed requests and notifies Alert Channels.

■ **Production Use Case:** During peak hours, when OpenAI returns HTTP 429 Too Many Requests, traffic automatically pivots to AWS Bedrock with zero disruption.

Pricing Overrides — Pricing Overrides (Custom Enterprise Rates & Discounts)

Route: </workspace/custom-pricing/overrides>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Special corporate discount allathu wholesale negotiated price contract mathiri.
- **Vilakkam:** OpenAI website-la 1 million token-ku \$5 nu irunthaalum, unga company-ku special enterprise discount allathu committed spend agreement iruntha, antha custom rate-ah inga enter pannikalam. Athuku thagundha mathiri unga internal dashboard-la exact real spend calculate aagum.
- **Business Value:** 100% accurate financial chargeback and departmental billing based on actual negotiated contracts.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Multi-tier hierarchical pricing precedence engine: (1) Virtual Key Override -> (2) User/Team Override -> (3) Provider Global Override -> (4) Default Provider Catalog. Calculates prompt tokens, cached input tokens, output tokens, reasoning tokens, and fixed invocation fees.
- **Endpoints:** GET `/api/v1/pricing/overrides`, POST `/api/v1/pricing/overrides`, PUT `/api/v1/pricing/overrides/{id}`, DELETE `/api/v1/pricing/overrides/{id}`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Add Pricing Override Button
- Scope Filter (Global, Virtual Key, Team)
- Search Input

2. Mathiya Koorugal & Tables (Tabs & Views):

- Pricing Overrides Table: Scope Badge, Target Entity, Provider, Model, Input Cost \$/1M, Cached Input Cost, Output Cost \$/1M, Request Fee, Actions

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- PricingOverrideSheet: Scope, Provider, Model, and PricingFieldSelector inputs for granular token pricing

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Finance administrators.
- **Iyakkum Koorugal:** Feeds real-time spend calculations into Dashboard, LLM Logs, and Budgets & Limits.

■ **Production Use Case:** Applying an AWS 20% committed spend discount to Bedrock Claude models so internal reporting matches cloud invoices.

Model Settings — Model Settings (Master Catalog Sync & Recursion Guard)

Route: `/workspace/custom-pricing`

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Global system clock & master registry synchronizer mathiri.
- **Vilakkam:** Market-la puthu puthu AI models varum pothu, athoda official pricing automatic-ah download aagi update aaga thevaiyana master website link, sync frequency (24 hours), mathum routing rules-la infinite loop vanthu server hang aagidama thadukkura max depth settings configure pandra edham.
- **Business Value:** Zero-maintenance operations; automated price drop ingestion without code deployments.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Core configuration manager in PostgreSQL (``framework_config``, ``client_config``). Orchestrates asynchronous cron workers to fetch remote pricing datasheets (CSV/JSON). Enforces ``routing_chain_max_depth`` to prevent circular fallback loops.
- **Endpoints:** GET `/api/v1/config/core`, PUT `/api/v1/config/core`, POST `/api/v1/config/pricing/sync`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Header Title
- Dirty State Tracker highlighting unsaved changes

2. Mathiya Koorugal & Tables (Tabs & Views):

- `ModelSettingsView`: Pricing Datasheet URL, Pricing Sync Interval (Hours), Model Parameters URL, Routing Chain Max Depth (slider/input)

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- Force Pricing Sync Now Button (with loading spinner)
- Save Configuration Button

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Admin inputs and remote pricing feeds.
- **Iyakkum Koorugal:** Updates foundational pricing for Model Catalog and prevents routing recursion loops.

■ **Production Use Case:** Instantly ingesting OpenAI's latest price reduction across 30 enterprise models with a single click of 'Force Pricing Sync Now'.

Pillar: MCP Gateway & Tool Execution (MCP Gateway (Model Context Protocol & Agent Tools))

Autonomous AI Agents company-oda internal databases, APIs, codebases, and external SaaS tools kooda interact panna thevaiyana standardized gateway.

MCP Catalog — MCP Catalog (Tool Registry & App Store)

Route: `/workspace/mcp-registry`

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** App Store allathu Google Play Store mathiri.
- **Vilakkam:** AI Agents use panna koodiya ellam certified tools (GitHub MCP, PostgreSQL MCP, Slack MCP, Google Drive MCP, Puppeteer Web Scraper) list irukkura marketplace. Single click-la puthu tool install pannikalam.
- **Business Value:** Fast-tracks AI agent deployment; verified secure tool marketplace; zero boilerplate integration.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Curated MCP registry client. Fetches certified tool schemas, version tags, environment variable templates, and execution commands (stdio binary or Docker container) from central repositories.
- **Endpoints:** GET `/api/v1/mcp/registry`, POST `/api/v1/mcp/registry/install`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Search Catalog Input
- Category Filter (Databases, Version Control, Communication, DevTools)
- Installed vs Available Filter

2. Mathiya Koorugal & Tables (Tabs & Views):

- Tool Cards Grid: Icon, Tool Name, Description, Verified Badge, Version, 'Install' button

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- InstallToolDialog: Environment variable inputs (API keys, connection strings) and transport mode selector

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** External MCP package repositories.
- **Iyakkum Koorugal:** Deploys tools into Installed Servers.

■ **Production Use Case:** Installing the PostgreSQL MCP tool to let internal analytics AI query corporate sales data safely.

Installed Servers — Installed Servers (Active MCP Instances & Health)

Route: `/workspace/mcp-gateway`

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Computer-la install aana software apps dashboard mathiri.
- **Vilakkam:** Company-kulla ippo active-ah run aagikittu irukura MCP tool servers list. Entha server online-la irukku, ethu crashed, ethana tools antha server tharuthu nu paathu restart allathu configure pandra idham.
- **Business Value:** Tool availability monitoring; instant recovery from crashed tool daemons; lifecycle control.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** MCP Server lifecycle daemon supervisor. Manages child OS processes (`stdio`) or HTTP Server-Sent Events (`SSE`) endpoints. Performs automatic process health checks, restarts crashed servers, and registers exported tool schemas.
- **Endpoints:** GET `/api/v1/mcp/servers`, POST `/api/v1/mcp/servers/{id}/restart`, DELETE `/api/v1/mcp/servers/{id}`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Add Server Button
- Search Servers
- Health Filter (Running, Degraded, Stopped)

2. Mathiya Koorugal & Tables (Tabs & Views):

- Installed Servers Table: Server Name, Transport (stdio/SSE), Status Badge, Tool Count, Memory Usage, Actions (Restart, Edit, Delete)

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- ServerConfigSheet: Command arguments, working directory, environment variables, timeout limits

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** MCP Catalog installations.
- **Iyakkum Koorugal:** Exposes tools to Tool Groups and logs calls to MCP Logs.

■ **Production Use Case:** Restarting the GitHub MCP server daemon after generating a new personal access token.

MCP Sessions — Sessions (Live Stateful Agent Connections)

Route: </workspace/mcp-sessions>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Active phone calls allathu live chat sessions list mathiri.
- **Vilakkam:** Ippo entha entha AI agents live-ah company tools kooda connect pannu vela senjukittu irukku nu kaatara active connection monitor. Thevaillatha session-ah force-ah terminate pannalam.
- **Business Value:** Prevents runaway autonomous agents; immediate containment of malfunctioning automation scripts.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Stateful session tracker for SSE and WebSocket MCP connections. Tracks session ID, client IP, connected agent identity, active context windows, and tool execution lock state.
- **Endpoints:** GET `/api/v1/mcp/sessions`, DELETE `/api/v1/mcp/sessions/{id}`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Active Sessions Count Badge
- Kill All Sessions Emergency Button

2. Mathiya Koorugal & Tables (Tabs & Views):

- Sessions Table: Session ID, Connected Agent/User, Duration, Tools Invoked Count, Memory State, Actions (Terminate)

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- Session Trace Modal: Live stream of JSON-RPC tool frames exchanged in this session

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Live incoming agent connections.
- **Iyakkum Koorugal:** Can terminate active sessions to protect downstream infrastructure.

■ **Production Use Case:** Killing an agent session that entered an infinite loop repeatedly calling the Slack notification tool.

Tool Groups — Tool Groups (Permission Bundles & Tool Access Control)

Route: </workspace/mcp-tool-groups>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Security clearance badge groups mathiri (Junior dev-ku read-only tools, Senior dev-ku write tools).
- **Vilakkam:** Multiple tools-ah onna sethu bundle pannalam. Example: 'Read-Only Tools' bundle (view database, read files) allathu 'Production Tools' bundle (write database, send email). Virtual key-ku etha mathiri intha groups-ah assign pannikalam.
- **Business Value:** Least-privilege security for autonomous AI; prevents AI from executing dangerous destructive commands.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Access control abstraction layer over MCP tool schemas. Maps Virtual Keys to permitted Tool Groups. Gateway filters out unpermitted tool definitions from the LLM prompt's system message dynamically.
- **Endpoints:** GET `/api/v1/mcp/tool-groups`, POST `/api/v1/mcp/tool-groups`, PATCH `/api/v1/mcp/tool-groups/{id}`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Create Tool Group Button
- Search Groups

2. Mathiya Koorugal & Tables (Tabs & Views):

- Tool Groups Cards: Group Name, Description, Included Tools count, Assigned Virtual Keys count, Actions

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- ToolGroupSheet: Group Name, Description, Interactive Tool Selector Checkboxes across all installed servers

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Installed Servers tools.
- **Iyakkum Koorugal:** Binds to Virtual Keys and Access Profiles.

■ **Production Use Case:** Restricting an intern's chatbot to only search internal documentation while preventing it from modifying the production database.

Auth Config — Auth Config (MCP Security & Client Tokens)

Route: </workspace/mcp-auth-config>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Security passport & digital signature verification mathiri.
- **Vilakkam:** External agents and client software MCP Gateway-kulla connect aaga thevaiyana OAuth 2.0, Bearer token, and mutual TLS authentication settings configure pandra edham.
- **Business Value:** Prevents unauthorized agents from connecting to internal enterprise tools.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** MCP Gateway authentication middleware. Validates incoming connection credentials against OAuth 2.0 introspection endpoints, API keys, or mTLS client certificates before allowing tool execution.
- **Endpoints:** GET </api/v1/mcp/auth-config>, PUT </api/v1/mcp/auth-config>

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Master Auth Enforcement Toggle
- Save Auth Config Button

2. Mathiya Koorugal & Tables (Tabs & Views):

- Auth Settings Form: Allowed Auth Types (Bearer, mTLS, OAuth2), Token Expiration Duration, Whitelisted Client IPs

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- Client Token Generator: Create dedicated secure connection tokens for trusted agent processes

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Security administrators.
- **Iyakkum Koorugal:** Authorizes or rejects incoming connections to MCP Gateway.

■ **Production Use Case:** Requiring mutual TLS certificates for all agent connections originating from external cloud environments.

Gateway Settings — Gateway Settings (MCP Engine Timeouts & Network Ports)

Route: </workspace/mcp-settings>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Factory master electricity & network switch settings mathiri.
- **Vilakkam:** MCP Gateway-oda core engine settings: Oru tool call maximum ethana seconds run aagalam (timeout), maximum payload size evlo irukkalam, background worker threads ethana nu configure pandra control panel.
- **Business Value:** Prevents slow tools from hanging server resources; guarantees gateway stability.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Core runtime configuration for the MCP Gateway subsystem. Manages maximum concurrent sessions, global execution timeouts, SSE heartbeat intervals, and payload size limits.
- **Endpoints:** GET </api/v1/mcp/settings>, PUT </api/v1/mcp/settings>

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Header Title
- Save Settings Button

2. Mathiya Koorugal & Tables (Tabs & Views):

- Settings Form: Global Tool Timeout ms, Max Payload Size MB, Max Concurrent Sessions, SSE Heartbeat Seconds

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- Reset to Recommended Defaults Button

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** System Administrators.
- **Iyakkum Koorugal:** Governs the execution environment of all Installed Servers and Sessions.

■ **Production Use Case:** Setting a strict 15-second timeout on all tool calls so a slow database query doesn't freeze the AI user experience.

Pillar: Prompt Management & Playground (Prompt Management (Prompt Repo, Skills & Chat Playground))

Enterprise prompt templates version control, reusable agent skills, and multi-model testing playground.

Prompt Repo — Prompt Repo (Version Controlled Prompt Templates)

Route: </workspace/prompt-repo>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Software code-kaga irukkura GitHub repository mathiri, prompts-kaga irukkura Git repository.
- **Vilakkam:** Developers application-la use pandra prompt templates-ah hardcode pannama, inga central-ah store pannu version control (v1, v2, v3) pannalam. Code redeploy pannama prompt-ah change pannu A/B test pannalam.
- **Business Value:** Decouples prompt engineering from backend deployments; enables non-technical domain experts to optimize prompts safely.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Prompt template versioning engine with variable interpolation (`{{user\_name}}`, `{{account\_id}}`). Supports commit messages, semantic diffing, rollback to previous versions, and production release tagging.
- **Endpoints:** GET `/api/v1/prompts`, POST `/api/v1/prompts`, GET `/api/v1/prompts/{id}/versions`, POST `/api/v1/prompts/{id}/deploy`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Create Prompt Button
- Search Prompts
- Tag Filter (Sales, Support, Code, Legal)

2. Mathiya Koorugal & Tables (Tabs & Views):

- Prompts Table: Prompt Name, Latest Version, Production Tag, Variables count, Last Updated, Actions
- Visual Diff Viewer: Side-by-side comparison of v1 vs v2

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- PromptEditorSheet: Template textarea, Variable detector, Model parameter settings (temperature, top_p)

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Prompt engineers and developers.
- **Iyakkum Koorugal:** Supplies production templates to client apps via FastHTTP Proxy and Guardrail Rules.

■ **Production Use Case:** Updating the company's customer support prompt to include holiday return policies instantly without touching backend server code.

Skills Repo — Skills Repo (Reusable Agent Competencies & Knowledge Packs)

Route: </workspace/skills-repo>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Specialized employee training certificates & skills handbook mathiri.
- **Vilakkam:** AI Agent seiya koodiya complex multi-step workflows-ah 'Skill'-ah define pannu store pannalam (e.g. 'Generate SQL Query', 'Summarize Financial PDF', 'Draft Legal Contract'). Entha agent venumnaalum intha skill-ah borrow pannikalam.
- **Business Value:** Massive reuse of engineered AI workflows across multiple company applications.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Modular skill definition registry. Packages system instructions, few-shot demonstration examples, required MCP tool bindings, and validation schemas into an executable skill bundle.
- **Endpoints:** GET `/api/v1/skills`, POST `/api/v1/skills`, PATCH `/api/v1/skills/{id}`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Create Skill Button
- Search Skills
- Category Filter

2. Mathiya Koorugal & Tables (Tabs & Views):

- Skills Grid: Skill Name, Version, Attached Tools count, Few-Shot Examples count, Actions Menu

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- SkillBuilderSheet: System instructions, Required tool bindings, Demonstration input/output pairs

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** AI Engineers.
- **Iyakkum Koorugal:** Enriches agent context during chat and automated agent workflows.

■ **Production Use Case:** Creating a standard 'Summarize Earnings Call' skill that automatically invokes financial tools and formats outputs consistently.

Chat / Playground — Playground (Multi-Model Testing & Comparison Sandbox)

Route: </workspace/chat>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Test drive track mathiri (Car vaangurathukku munnadi test drive panni pakkura mathiri).
- **Vilakkam:** Different AI models-ah (GPT-4o vs Claude 3.5 Sonnet vs Gemini 1.5) side-by-side compare panni, speed, answer quality, and cost-ah live-ah test panni pakkura interactive testing playground.
- **Business Value:** Fast experimentation; empirical model evaluation before committing to production deployments.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Interactive multi-model streaming sandbox. Dispatches concurrent SSE requests to multiple providers simultaneously. Displays real-time Time-To-First-Token (TTFT), tokens per second (TPS), and dollar cost per response.
- **Endpoints:** POST `/api/v1/chat/completions`, GET `/api/v1/models`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Model Selector Dropdowns (Multi-Model Split Screen)
- Temperature & Max Tokens Sliders
- System Prompt Input

2. Mathiya Koorugal & Tables (Tabs & Views):

- Split-Screen Chat View: Compare Model A vs Model B side-by-side in real time
- Live Latency & Cost Counter under each bubble

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- Export Chat to Prompt Repo Button
- Parameter Configuration Sheet

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Direct user prompts and templates from Prompt Repo.
- **Iyakkum Koorugal:** Executes calls across Model Providers and displays telemetry.

■ **Production Use Case:** Testing whether Claude 3.5 Sonnet writes better SQL queries than GPT-4o on proprietary company database schemas.

Pillar: Plugins & Extensibility (Plugins (Gateway Extensions & Custom Logic))

Gateway functionality-ah expand panna custom logic, billing webhooks, and proprietary headers inject pandra extensible engine.

Plugins — Plugins (Gateway Extensions & Lifecycle Hooks)

Route: </workspace/plugins>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Chrome extensions allathu WordPress plugins mathiri.
- **Vilakkam:** Gateway core code-ah maathaama, ungalukku thevaiyana custom rules (e.g. proprietary token counter, custom headers, billing webhooks) add panna thevaiyana extensible plugin store.
- **Business Value:** Zero friction integration of custom enterprise logic and legal policies directly into the proxy request pipeline.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** WebAssembly (Wasm) and Go-based lifecycle interceptor pipeline. Hooks into `pre-request`, `post-request`, `response-streaming`, and `error-handling` execution phases with ordered priority.
- **Endpoints:** GET `/api/v1/plugins`, POST `/api/v1/plugins`, PUT `/api/v1/plugins/sequence`, DELETE `/api/v1/plugins/{id}`

■■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Add New Plugin Button
- Plugin Sequence Button (ListOrdered)

2. Mathiya Koorugal & Tables (Tabs & Views):

- Plugins Sidebar with status badges
- PluginsView: Name, Description, Version, Schema, Active Toggle

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- AddNewPluginSheet: Name, Code/URL, Hook selectors
- PluginSequenceSheet: Visual drag-and-drop execution ordering

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** FastHTTP Proxy raw request.
- **Iyakkum Koorugal:** Enriches headers, mutates payloads, and logs traces to LLM Logs.

■ **Production Use Case:** Injecting an enterprise compliance plugin that validates legal disclaimers on every financial advice prompt.

Pillar: Governance & Identity (Governance (Identity, RBAC, Virtual Keys & Security))

Multi-tenant enterprise hierarchy, synthetic Virtual Keys, automated SCIM provisioning, RBAC permissions, and immutable audit trails.

Virtual Keys — Virtual Keys (Proxy API Keys & Granular Quotas)

Route: </workspace/governance/virtual-keys>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Bank master account-ku sub-debit cards mathiri (ovvoru card-kum thani limit).
- **Vilakkam:** Master OpenAI keys-ah developers kitta direct-ah tharama, dummy Virtual Key create pannu tharuvom. Monthly \$100 limit, specific models permission potruppom. Key leak aanaalum single click-la revoke pannikalam, real keys safe-ah irukkum.
- **Business Value:** Zero credential leakage risk; 100% cost attribution by application; instant security revocation.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** High-throughput cryptographic API key gateway middleware. Keys are generated with cryptographically secure tokens ('uf-key-...') and stored as SHA-256 hashes. FastHTTP middleware validates tokens in < 1ms, checks budget balances, and enforces RPM/TPM limits.
- **Endpoints:** GET </api/v1/governance/virtual-keys>, POST </api/v1/governance/virtual-keys>, POST </api/v1/governance/virtual-keys/rotate>

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Create Virtual Key Button
- Bulk Rotate Keys Button
- Search & Filter Bar

2. Mathiya Koorugal & Tables (Tabs & Views):

- Virtual Keys Table: Name, Masked Secret (copy/eye toggle), Team, Customer, BudgetDisplay progress bar, RateLimitDisplay badge, Status switch, Actions

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- VirtualKeySheet: Team, Access Profile, Budget Cap \$, Rate Limits, Model Allowlist, Expiration Date
- One-Time Secret Reveal Modal

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Teams, Customers, and Access Profiles.
- **Iyakkum Koorugal:** Authorizes incoming requests in FastHTTP Proxy and tracks spend into LLM Logs.

■ **Production Use Case:** Issuing a virtual key with a \$150/month cap and Gemini-only access to an external vendor development squad.

Users — Users (Platform User Directory & Identity Management)

Route: </workspace/governance/users>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Company employee directory & staff ID register mathiri.
- **Vilakkam:** UnifAI platform-la account irukkura ellam developers, team leads, finance managers, and admins list. Yaar yaar enna email, enna role, entha team-la irukaanga nu manage pandra central user directory.
- **Business Value:** Centralized identity governance; seamless onboarding and automated access termination when staff leave.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** User identity and authentication manager integrated with enterprise SSO (SAML 2.0 / OIDC) and automated SCIM provisioning. Sessions are signed via JWTs with mandatory MFA support.
- **Endpoints:** GET </api/v1/governance/users>, POST </api/v1/governance/users/invite>, PATCH </api/v1/governance/users/{id}>

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Invite User Button
- Search Users
- Filter by Role / Status

2. Mathiya Koorugal & Tables (Tabs & Views):

- Users Table: Avatar, Name, Email, Assigned RBAC Role badge, Team Memberships, Last Active, Status badge, Actions

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- InviteUserDialog: Email, Full Name, Initial Role, Assigned Teams
- EditUserSheet: Role adjust, Suspend user toggle

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** SCIM / SSO or Admin invites.
- **Iyakkum Koorugal:** Binds to Teams, Virtual Keys, and RBAC roles; actions logged to Audit Logs.

■ **Production Use Case:** Inviting 25 software engineers to the platform with Member-level permissions assigned to the Search Squad.

Teams — Teams (Departmental Squads & Shared Resource Pools)

Route: </workspace/governance/teams>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Company project squads allathu departments mathiri (e.g. Mobile Squad, Search Team).
- **Vilakkam:** Developers-ah group pannu team-ah pirikalam. Oru team-ku common monthly budget, shared virtual keys, and collaborative model access kudukalam.
- **Business Value:** Departmental accountability; team-level AI spend tracking for CFO budget reviews.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Multi-tenant team resource isolation layer. Aggregates usage across all member users and attached virtual keys. Enforces team-level budget caps in PostgreSQL.
- **Endpoints:** GET `/api/v1/governance/teams`, POST `/api/v1/governance/teams`, PATCH `/api/v1/governance/teams/{id}`

■■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Create Team Button
- Search Teams
- Business Unit Filter

2. Mathiya Koorugal & Tables (Tabs & Views):

- Teams Table: Team Name, Description, Members count badge, Virtual Keys count, Monthly Budget Progress Bar, Parent Business Unit

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- CreateTeamDialog: Name, Description, Parent Business Unit, Monthly Budget \$, Team Lead
- ManageMembersSheet

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Belongs to a Business Unit; groups Users.
- **Iyakkum Koorugal:** Owns shared Virtual Keys; feeds spend into Dashboard.

■ **Production Use Case:** Allocating a shared \$4,000 monthly budget to the 'Mobile App Team' with 15 engineers.

Business Units — Business Units (Enterprise Divisions & P&L; Centers)

Route: </workspace/governance/business-units>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Corporate conglomerate divisions mathiri (Retail Division, Banking Division).
- **Vilakkam:** Teams-ku mela irukura periyai enterprise division. CFO and executive leadership company-oda multiple business units-kulla AI budget■■■■ pannavum, P&L; tracking seiyavum use aagum.
- **Business Value:** High-level corporate financial governance; automated chargebacks across major business divisions.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Top-level container in the multi-tenant governance tree (`BusinessUnit -> Teams -> Users -> Virtual Keys`). Aggregates financial consumption for corporate ERP systems (SAP, NetSuite).
- **Endpoints:** GET `/api/v1/governance/business-units`, POST `/api/v1/governance/business-units`

■■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Create Business Unit Button
- Search Divisions

2. Mathiya Koorugal & Tables (Tabs & Views):

- Business Units Table: Unit Name, Cost Code, Total Sub-Teams count, Aggregate Spend \$, Head of Division

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- BusinessUnitSheet: Unit Name, Cost Center Code, Owner Email, Quarterly Budget Cap \$

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Executive management.
- **Iyakkum Koorugal:** Contains multiple Teams; feeds top-level cost analytics into Executive Dashboards.

■ **Production Use Case:** Tracking that the 'Healthcare Services' division consumed \$60,000 in AI tokens across its 10 engineering teams.

Customers — Customers (B2B Client Tenants & External Accounts)

Route: </workspace/governance/customers>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** SaaS B2B client accounts mathiri.
- **Vilakkam:** Neenga unga software-ah vera external clients-ku AI features vachu SaaS product-ah vikkiringa na, ovvoru client-aiyum 'Customer'-ah add pannu avanga usage-ku bill podalam. Oru customer innoru customer-oda data-vai paarka mudiyathu.
- **Business Value:** Monetize AI applications; seamless B2B client isolation; automated billing invoices based on client consumption.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** External tenant isolation boundary. Associates virtual keys with external customer identifiers (`customer\_id`). Enforces hard tenant isolation policies in PostgreSQL using Row-Level Security (RLS).
- **Endpoints:** GET </api/v1/governance/customers>, POST </api/v1/governance/customers>

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Create Customer Button
- Search & Filter Bar

2. Mathiya Koorugal & Tables (Tabs & Views):

- Customers Table: Customer Name, Status badge, Assigned Teams, Virtual Keys count, Cumulative Spend \$, Tier, Actions

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- CustomerSheet: Customer Name, External Account ID, Contact Email, Assigned Pricing Tier, Hard Spend Quota

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Sales/CRM sync.
- **Iyakkum Koorugal:** Scopes Virtual Keys and filters LLM Logs for customer-specific billing.

■ **Production Use Case:** Giving 50 B2B enterprise clients their own dedicated virtual keys with strictly metered billing for your AI legal review software.

User Provisioning (SCIM) — User Provisioning / SCIM (Automated Enterprise SSO Lifecycle)

Route: </workspace/scim>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Automatic company ID card issuing & revoking machine mathiri.
- **Vilakkam:** Okta, Microsoft Entra ID / Azure AD, Keycloak-la pudhu aal sertha automatic-ah UnifAI-layum login create aagum. Employee velaiya vittu ponavudane Okta-la offboard pannina, automatic-ah UnifAI access cut aagidum!
- **Business Value:** Zero orphan accounts; 100% SOC-2 compliance for employee lifecycle security; automated enterprise IT operations.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Standard SCIM v2.0 (RFC 7643 & RFC 7644) server. Handles `/scim/v2/Users` and `/scim/v2/Groups` with Bearer Token auth. Automatically creates, updates, deactivates users, and maps directory groups to Teams.
- **Endpoints:** GET </scim/v2/Users>, POST </scim/v2/Users>, GET </scim/v2/Groups>, GET </api/v1/scim/config>

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- SCIM Master Enable Switch
- IdP Provider Selector (Okta, Entra ID, Keycloak)

2. Mathiya Koorugal & Tables (Tabs & Views):

- SCIM Endpoints Card: Base URL, Users endpoint, ServiceProviderConfig (with copy buttons)
- Group Mapping Card

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- Bearer Token Generator with copy button and rotation warning
- Save SCIM Config Button

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** External Identity Providers (Okta, Entra ID).
- **Iyakkum Koorugal:** Provisions and deprovisions Users and synchronizes Teams.

■ **Production Use Case:** Auto-provisioning UnifAI accounts for 1,500 corporate employees via Okta SCIM sync.

Roles & Permissions (RBAC) — Roles & Permissions (RBAC Security & Access Matrix)

Route: </workspace/governance/rbac>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Office security access badge permissions mathiri.
- **Vilakkam:** Yaaru platform-la enna seiyalaam nu kattu paduthum access control matrix. Super Admin-ku full control; Developer-ku API key create panna mattum permission; Finance-ku logs and billing charts mattum paarka permission (read-only).
- **Business Value:** Least-privilege security principle. Prevents accidental deletions and unauthorized key tampering.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Declarative Role-Based Access Control (RBAC) engine in Go and React. Enforces fine-grained resource-action pairs ('Resource: ModelProvider, VirtualKeys, RoutingRules, Settings, AuditLogs, Plugins' × 'Action: View, Create, Update, Delete').
- **Endpoints:** GET `/api/v1/governance/rbac/roles`, POST `/api/v1/governance/rbac/roles`, GET `/api/v1/governance/rbac/permissions`

■■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Create Custom Role Button
- Role Filter (System vs Custom)

2. Mathiya Koorugal & Tables (Tabs & Views):

- Roles Table: Role Name, Description, Assigned Users count, Type badge, Actions
- RBAC Permission Matrix Grid: Resources on rows, Actions on columns with checkboxes

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- RoleBuilderSheet: Role Name, Description, Permission checkboxes
- AssignUsersModal

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Assigned to Users and Teams.
- **Iyakkum Koorugal:** Controls authorization on every API and UI view.

■ **Production Use Case:** Creating a 'Finance Auditor' role that can View Dashboard and View LLM Logs, but cannot create or delete any keys or models.

Access Profiles — Access Profiles (Reusable Policy Bundles & Guardrail Packs)

Route: </workspace/governance/access-profiles>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Pre-packaged security passport allathu policy bundle mathiri.
- **Vilakkam:** Oru standard policy template create panni vachikalam (e.g. 'Intern Profile': GPT-4o-mini mattum allow panni, \$50 budget limit, PII redaction ON). Pudhu virtual key create pannum pothu intha profile-ah select panna ella security rules-um automatic-ah apply aagidum!
- **Business Value:** Administrative time savings; eliminates configuration mistakes; guarantees that no virtual key goes live without security guardrails.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Reusable policy entity bundling Allowed Models, Allowed Providers, Budget Caps, Rate Limits (RPM/TPM), MCP Tool Groups, and Guardrails. Attached to Virtual Keys via relational join.
- **Endpoints:** GET `/api/v1/governance/access-profiles`, POST `/api/v1/governance/access-profiles`, POST `/api/v1/governance/access-profiles/{id}/clone`

■■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Create Access Profile Button
- Search & Tag Filter

2. Mathiya Koorugal & Tables (Tabs & Views):

- Access Profiles Table: Profile Name, Tags badges, Allowed Models, Attached Keys count, Budget/Rate limit summary, Actions

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- AccessProfileDialog: Allowed Models multiselect, Budget Caps \$, RPM/TPM Rate limits, MCP Tool Client pickers
- Quick Clone Button

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Security administrators.
- **Iyakkum Koorugal:** Governs Virtual Keys behavior and restricts Model Providers and MCP tools.

■ **Production Use Case:** Attaching a single 'Customer Chatbot Profile' to 30 virtual keys, guaranteeing PII redaction and 60 RPM limits across all 30 keys simultaneously.

Audit Logs — Audit Logs (Tamper-Proof Compliance & Activity Trail)

Route: `/workspace/audit-logs`

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Bank locker CCTV camera & register note mathiri.
- **Vilakkam:** Platform-la yaaru yaaru entha nerathula enna maathinaanga nu maatha mudiyaatha (tamper-proof) pathivu. Yaarachum API key delete pannalaam, budget limit maathinaalo, pudhu routing rule pottaalo exact timestamp, user name, and client IP address-oda capture aagum.
- **Business Value:** 100% SOC-2, HIPAA, and GDPR audit readiness; insider threat detection; rapid forensic debugging.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Immutable, append-only security event audit trail. Captures Actor User ID, Action Type (CREATE, UPDATE, DELETE), Target Resource, Timestamp, Client IP address, User-Agent, and before/after JSON diffs in PostgreSQL.
- **Endpoints:** GET `/api/v1/audit-logs`, GET `/api/v1/audit-logs/{id}`, GET `/api/v1/audit-logs/export`

■■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Date Range Picker
- Actor Search
- Action Type Filter (CREATE, UPDATE, DELETE)
- Resource Filter
- Export Button (CSV/JSON)

2. Mathiya Koorugal & Tables (Tabs & Views):

- Audit Logs Table: Timestamp, Actor Name & Email, Action Badge, Resource Type, Target ID, Client IP, Details Button

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- Audit Detail Drawer: Displays complete before-and-after JSON delta diff highlighting modified fields

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Intercepts administrative mutations across all Governance, Models, and Observability features.
- **Iyakkum Koorugal:** Feeds security alerts and provides compliance proof for auditors.

■ **Production Use Case:** Demonstrating to external SOC-2 auditors exactly who rotated the production AWS Bedrock credentials on July 10th.

Pillar: Security Guardrails (Guardrails (Content Safety & Attack Prevention))

Prompt injection prevention, PII redaction, toxicity filtering, and multi-engine security scanners.

Guardrail Rules — Guardrail Rules (AI Safety Policies & Screening)

Route: </workspace/guardrails/configuration>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Airport baggage scanner & security checkpoint mathiri.
- **Vilakkam:** Customer allathu employee AI kitta bad words, company secrets, credit card numbers, allathu jailbreak / prompt injection attacks seiyatha mathiri thadukkura security rules. Rules meera patta udane request-ah block pannum allathu PII data-vai asterisk (*) pottu mask pannum.
- **Business Value:** Zero data leakage (PII/Secrets); regulatory compliance (HIPAA/GDPR); brand reputation protection.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Pre-request and post-completion content security policy engine in Go. Evaluates prompt scope ('all', specific Prompt Repo IDs, or custom CEL expressions). Rejects violations with HTTP 400/403 with verdict codes ('PII_DETECTED', 'PROMPT_INJECTION').
- **Endpoints:** GET /api/v1/guardrails/config, PUT /api/v1/guardrails/config, POST /api/v1/guardrails/test

■■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Header with ShieldAlert Icon
- Add Rule Button
- Active Rules Count Badge

2. Mathiya Koorugal & Tables (Tabs & Views):

- Guardrail Rules Table: Rule Name, Active Toggle, Prompt Scope, Linked Providers, Evaluation Action, Actions

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- GuardrailRuleDialog: Rule Name, Linked Providers multiselect, Prompt Scope Selector (All, Selected Prompts, Custom CEL), CEL Editor

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** FastHTTP Proxy and Prompt Repository.
- **Iyakkum Koorugal:** Blocks attacks, redacts sensitive text, triggers Alert Channels, and logs incidents to LLM Logs.

■ **Production Use Case:** Preventing chatbot users from extracting system prompts or submitting SQL injection payloads into AI text fields.

Guardrail Providers — Guardrail Providers (Security Engines & Regex Scanners)

Route: </workspace/guardrails/providers>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Security inspection scanner machinery brands mathiri.
- **Vilakkam:** Security rules run aagurathukku thevaiyana scanning engines (Presidio PII Engine, Llama-Guard Toxicity Engine, Lakera Prompt Injection Scanner, AWS Bedrock Guardrails, allathu Custom Regex Pattern matching engines) configure pandra edham.
- **Business Value:** Customizable data defense; multi-layered scanning capability; adaptability to enterprise-specific regex patterns.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Multi-engine security scanner registry. Supports local RE2 regex matching, lightweight embeddings classifiers, and external gRPC/HTTP safety microservices (Microsoft Presidio, Meta Llama-Guard). Concurrent Go worker execution (< 15ms).
- **Endpoints:** GET /api/v1/guardrails/config, PUT /api/v1/guardrails/config

■■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Add Provider Button
- Navigation Tabs (Rules vs Providers)

2. Mathiya Koorugal & Tables (Tabs & Views):

- Guardrail Providers Table: Provider ID, Policy Name, Engine Type (Regex/Presidio/Bedrock), Pattern Count badge, Linked Rules count badge, Actions

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- GuardrailProviderDialog: Provider ID, Policy Name, Engine Type, Multi-line Regex Patterns textarea, Pattern Syntax Validator

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** DevSecOps engineers.
- **Iyakkum Koorugal:** Supplies scanning algorithms and regex patterns to Guardrail Rules.

■ **Production Use Case:** Configuring custom regex patterns to detect and mask proprietary internal project code names before prompts reach public models.

Pillar: High Availability & Infrastructure (Infrastructure (Cluster Mesh, Alerts & OAuth))

Multi-region cluster synchronization, Slack/PagerDuty alert channels, and third-party OAuth access grants.

Cluster Config — Cluster Config (Distributed Mesh & High-Availability Sync)

Route: </workspace/cluster>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Multi-aircraft flight formation allathu multi-branch bank network mathiri.
- **Vilakkam:** Multiple UnifAI servers (USA, Europe, India regions) run aagum pothu, antha ellam servers-um onnukonnu pesikittu, rate limits, virtual key balances, and circuit breaker states instant-ah sync aaguratha uruthi seiyura master cluster control panel.
- **Business Value:** Zero single point of failure; multi-region active-active deployment; global consistency.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Distributed Mesh / Gossip protocol (HashiCorp Memberlist / Serf). Nodes communicate via TCP/UDP peer gossip over port 7946. Synchronizes in-memory rate-limit buckets and circuit breaker states (< 50ms) across regions.
- **Endpoints:** GET `/api/v1/cluster/config`, PUT `/api/v1/cluster/config`, GET `/api/v1/cluster/nodes`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Header with Network Icon
- Cluster Master Enable Switch

2. Mathiya Koorugal & Tables (Tabs & Views):

- Cluster Topology Card: Cluster Type (Mesh/Gossip), Region ID, Peer Seed List textarea (host:port, e.g. 10.0.0.12:7946)
- Node Info Card: Node ID, Bind Address, Role, Active Peers count

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- Peer Syntax Validator with inline warnings
- Save Configuration Button

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** DevOps / SRE teams.
- **Iyakkum Koorugal:** Synchronizes Virtual Key rate limits and Circuit Breaker states across all gateway pods.

■ **Production Use Case:** Deploying 10 UnifAI gateway pods in Kubernetes, ensuring a user's 60 RPM rate limit is enforced globally rather than allowing 600 requests.

Alert Channels — Alert Channels (Incident Notifications & Webhooks)

Route: </workspace/alert-channels>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Emergency fire alarm & automated SMS siren mathiri.
- **Vilakkam:** AI provider crash aanaalo, budget 90% reach aanaalo, allathu prompt injection attack nadanthalo, on-duty engineers-ku instant-ah Slack, PagerDuty, Microsoft Teams, allathu Email alert anuppura emergency notification hub.
- **Business Value:** Rapid incident response; minimizes downtime; prevents surprise cloud billing spikes.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Multi-channel incident notification dispatcher. Formats structured alerts into Slack webhook blocks, PagerDuty Events v2 JSON, or SMTP email alerts. Implements notification de-duplication and rate dampening.
- **Endpoints:** GET `/api/v1/alert-channels`, POST `/api/v1/alert-channels`, POST `/api/v1/alert-channels/{id}/test`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Add Alert Channel Button
- Channel Type Filter (Slack, PagerDuty, Email, Webhook)

2. Mathiya Koorugal & Tables (Tabs & Views):

- Alert Channels Table: Channel Name, Type badge, Webhook Endpoint, Subscribed Events badges, Health Status, Actions

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- AddChannelDialog: Channel Name, Type selector, Webhook URL, Event Subscriptions checkboxes, Send Test Notification button

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Triggered by Circuit Breaker trips, Budgets & Limits warnings, and Guardrail violations.
- **Iyakkum Koorugal:** Dispatches messages to corporate Slack and PagerDuty.

■ **Production Use Case:** Triggering a high-urgency PagerDuty incident when OpenAI fails and Circuit Breaker activates the backup Azure route.

OAuth Grants — OAuth Grants (Third-Party App Authorizations)

Route: `/workspace/oauth-grants`

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Mobile phone-la 'Allow app to access camera/contacts' permission screen mathiri.
- **Vilakkam:** External software and AI developer tools (Cursor IDE, Continue.dev, internal apps) UnifAI gateway-kulla login panni access kettu varum pothu, antha third-party app-ku permission grant pandra OAuth 2.0 authorization center.
- **Business Value:** Standardized, secure enterprise developer tool integration without sharing permanent secret keys.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** OAuth 2.0 / OIDC Authorization Server. Manages OAuth client IDs, client secrets, redirect URIs, authorization code exchanges, and refresh token rotation. Issues scoped access tokens.
- **Endpoints:** GET `/api/v1/oauth/grants`, POST `/api/v1/oauth/clients`, DELETE `/api/v1/oauth/grants/{id}`

■■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Register OAuth Client Button
- Search Grants

2. Mathiya Koorugal & Tables (Tabs & Views):

- Active Grants Table: Client App Name, User, Scopes Granted badges, Issued At, Expires At, Revoke Access Button

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- RegisterClientDialog: App Name, Redirect URIs, Allowed Grant Types, Requested Scopes

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** External developer tools (Cursor, LangChain apps).
- **Iyakkum Koorugal:** Issues scoped Virtual Key equivalent tokens for API access.

■ **Production Use Case:** Authorizing developer Cursor IDE instances to route requests through UnifAI via OAuth 2.0 login.

Pillar: Adaptive Routing & Load Balancing (Adaptive Routing (Dynamic Latency & Score-Based Balancing))

Multi-Armed Bandit real-time latency scoring, automated key weighting, and unhealthy route pruning.

Adaptive Routing Dashboard — Adaptive Routing Dashboard (Live Health Scoring & Traffic Steering)

Route: </workspace/adaptive-routing>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Google Maps Live Traffic Navigation mathiri.
- **Vilakkam:** Oru AI model (OpenAI GPT-4o) ippo slow aagiduchuna allathu error adikkithu na, system live-ah detect panni, automatically athe tharam konda innoru fast-ana provider-ku (Azure OpenAI allathu Anthropic) traffic-ah thiruppi vittu user-ku semma fast response tharum.
- **Business Value:** Guarantees lowest possible latency; automated self-healing routing during vendor slowdowns.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Multi-Armed Bandit (MAB) reinforcement learning load balancer. Evaluates provider directions and individual API key routes based on live p50/p95 latency and error rates. Polled every 8 seconds (`pollingInterval: 8000ms`). Dynamically assigns routing weights.
- **Endpoints:** GET `/api/v1/load-balancer/routes`, PUT `/api/v1/load-balancer/config`, GET `/api/v1/load-balancer/metrics`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Header with Gauge Icon
- Settings Button deep-link

2. Mathiya Koorugal & Tables (Tabs & Views):

- 3 Summary Metric Cards: Load Balancer Switch, Live Scoring Strategy, Active Dynamic Routes
- Live Adaptive Routes Table: Provider/Direction, Model Name, Health Score (0-100 color badge), p50/p95 Latency ms, Error Rate %, Traffic Weight %, Status

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- 8-second background auto-refresh poller reflecting live shifting traffic weights

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** Latency and error telemetry from FastHTTP Proxy and Model Providers.
- **Iyakkum Koorugal:** Dynamically overrides Routing Rules traffic weights.

■ **Production Use Case:** When OpenAI latency spikes from 800ms to 4,500ms, adaptive routing automatically shifts 85% of traffic to Azure OpenAI maintaining 650ms p95 latency.

Adaptive Routing Settings — Adaptive Routing Settings (Load Balancing Policy & Pruning Rules)

Route: </workspace/adaptive-routing/settings>

■ Non-Tech Perspective (Eliya Vilakkam):

- **Uruvagam (Analogy):** Car cruise control & autopilot driving preference settings mathiri.
- **Vilakkam:** Adaptive load balancer eppadi nadanthukanum nu rules poduvom: Slow-ana vendor-ah automatic-ah ignore pannanuma? Oru vendor fail aana automatic-ah adutha vendor-ku reroute pannanuma? Multi-key balancing on pannanuma nu control pandra settings page.
- **Business Value:** Predictable routing behavior; fine-grained engineering control over automated failovers.

■ Tech Perspective (Technical Architecture):

- **Backend Architecture:** Load balancer configuration and policy persistence engine in PostgreSQL (`LoadBalancerConfig`). Controls direction selection algorithms, key-level round-robin weighting, failed direction rerouting, and fallback pruning.
- **Endpoints:** GET `/api/v1/load-balancer/config`, PUT `/api/v1/load-balancer/config`

■ Screen Layout & Interactive Elements (Thirai Koorugal):

1. Melpura Koorugal (Top Bar Controls):

- Header Title
- Back Button to Dashboard
- Save Settings Button

2. Mathiya Koorugal & Tables (Tabs & Views):

- Settings Card with 5 Policy Toggle Rows: (1) Enable adaptive load balancing, (2) Direction selection, (3) Route selection, (4) Reroute failed directions, (5) Prune failed fallbacks

3. Keezhpura Koorugal & Sheets (Bottom Elements & Slide-Overs):

- Save Configuration Button with instant toast confirmation

■ Connections & Structure Map (Inaippugal):

- **Data-vai Perumidam:** DevOps / Infrastructure leads.
- **Iyakkum Koorugal:** Governs the algorithmic behavior of Adaptive Routing Dashboard.

■ **Production Use Case:** Enabling 'Reroute failed directions' ensuring critical production apps never suffer timeouts from degraded upstream vendors.

3. Master Cross-Feature Interconnections Matrix

Source Feature	Connected To	Category	Data Flow & Trigger Action
Dashboard	LLM Logs	Observability	Visualizes real-time request volume, token usage, latencies, and costs.
LLM Logs	Connectors	Observability	Streams structured transaction events to Datadog, Kafka, and BigQuery.
MCP Logs	MCP Gateway	MCP	Records JSON-RPC arguments, tool execution traces, and stdout/stderr outputs.
Browser AI	Audit Logs	Security	Intercepts employee web AI paste events and logs confidential DLP violations.
Model Catalog	Routing Rules	Models	Provides verified model identifiers and context parameters for routing targets.
Model Providers	FastHTTP Proxy	Core	Supplies decrypted API keys, custom base URLs, and connection pools.
Budgets & Limits	Virtual Keys	Governance	Enforces strict dollar quotas and token rate limits on API keys.
Complexity Router	Routing Rules	Models	Scores query complexity (0-1) and assigns model tier (Simple to Reasoning).
Circuit Breaker	Alert Channels	Infrastructure	Triggers high-priority PagerDuty alerts when a primary provider fails.
Pricing Overrides	Dashboard & Billing	Finance	Calculates real dollar spend using negotiated corporate discount rates.
MCP Gateway	Tool Groups	MCP	Organizes certified tools into permission groups for Virtual Key assignment.
Prompt Repo	Guardrail Rules	Security	Links prompt templates to content screening and jailbreak detection rules.
Plugins	FastHTTP Proxy	Extensibility	Executes Wasm/Go pre-request, post-request, and streaming lifecycle hooks.
Virtual Keys	Access Profiles	Governance	Inherits model restrictions, budget limits, and MCP permissions.
SCIM	Users & Teams	Identity	Auto-provisions users and syncs directory groups from Okta and Entra ID.
RBAC	All UI & APIs	Security	Enforces declarative View/Create/Update/Delete permissions on every endpoint.
Audit Logs	All Features	Compliance	Immutably records every configuration change, key rotation, and admin action.
Guardrail Rules	Guardrail Providers	Security	Executes RE2 regex, Presidio, or Bedrock scanner engines on prompt text.
Cluster Config	Virtual Keys & Circuit	Infrastructure	Synchronizes rate-limit buckets and circuit trip states globally (:7946).
Adaptive Routing	Model Providers	Performance	Continuously scores provider latencies (ms) and shifts traffic dynamically.

4. Master Tech vs Non-Tech Comparative Matrix

Feature	Non-Tech View (Manager / CFO Parvai)	Tech View (DevOps / Architect Parvai)
Dashboard	AI spend evlo aaguthu, budget control-la irukka nu ore screen-la pakkalam.	WebSocket telemetry & atomic counters aggregating p95 latencies and TTFT into Postgres.
LLM Logs	Yaaru enna kelvi kettanga, AI enna bathil sonnathu nu passbook mathiri check pannalam.	Sub-millisecond PostgreSQL batch-insert ledger capturing raw payloads, tokens, and status codes.
MCP Logs	AI Agent company database-la enna vela senjithu nu inspect pannalam.	JSON-RPC 2.0 tool execution tracing across stdio and SSE agent transport protocols.
Browser AI	Employees company secrets-ah web ChatGPT-la copy-paste pannama thadukkalam.	Client-side mitmproxy daemon intercepting web AI pastes with Presidio DLP regexes.
Connectors	AI spend and logs automatic-ah namma corporate BigQuery/Datadog-ku poiduma?	Async streaming queue worker pushing events via OpenTelemetry, Kafka, and HTTP webhooks.
Model Catalog	Entha entha AI models namakku irukku? Ethu active-ah irukku nu menu card mathiri pakkalam.	FastHTTP model registry with capability filtering (vision, tool calling) and 24h usage metrics.
Model Providers	OpenAI, Anthropic, Bedrock API keys safe-ah connect aagi run aagutha?	Multi-vendor credential vault, FastHTTP client connection pools, and exponential retry backoffs.
Budgets & Limits	Excessive bill varama ovvoru team-kum monthly budget cap potu stop panna mudiyuma?	Atomic Redis/Postgres token bucket rate limiters evaluating multi-tiered limits with HTTP 429.
Complexity Router	Simple kelvikku cheap model, tough kelvikku matum costly model use panni 70% bill micham aagutha?	Heuristic multi-factor scoring engine categorizing queries into Simple, Med, Complex, Reasoning.
Circuit Breaker	OpenAI crash aanaal kooda customer-ku error theriyaama backup model-ku switch aaguma?	Distributed finite state machine (Closed/Open/Half-Open) monitoring response header signals.
MCP Catalog	AI agent-ku thevaiyana tools (GitHub, SQL, Slack) app store mathiri install pannalama?	Curated MCP registry client fetching validated schemas and execution configurations.
Prompt Repo	Prompt templates-ah code mathiri GitHub style-la version control panni release pannalama?	Prompt template engine with variable interpolation (<code>{{var}}</code>), semantic diffs, and release tags.
Plugins	Custom company rules and billing webhooks gateway-kulla plug-and-play-ah poda mudiyuma?	Wasm/Go lifecycle interceptor pipeline hooking into pre-request, post-request, and streaming chunks.
Virtual Keys	Real OpenAI master keys-ah tharama safe-ana dummy keys with budget limits tharalama?	Cryptographic SHA-256 hashed API tokens with sub-millisecond FastHTTP auth and rate limiters.
SCIM	Okta-la employee join aana automatic-ah account create aagi, leave aana cut aaguma?	RFC 7643/7644 SCIM v2.0 server handling automated user lifecycle and group synchronization.
RBAC	Admin-ku mattum full rights, mathavangalukku view-only permission pottu lock pannalama?	Declarative resource-action permission matrix evaluating gates across all UI views and REST APIs.
Access Profiles	Reusable standard security template create panni 20 keys-ku single click-la apply pannalama?	Bundled governance policy entity containing model allowlists, budget caps, rate limits, and MCP tools.
Audit Logs	Yaaru entha key-ah maathinaanga, eppo delete panninaanga nu maatha mudiyatha proof irukka?	Append-only immutable audit trail capturing actor, action, timestamp, IP, and before/after JSON diffs.
Guardrail Rules	Prompt injection, bad words, and company secrets AI kitta pogama thadukkuma?	CEL conditional content security engine scanning prompts/responses and returning 400/403 on violation.
Cluster Config	Multiple servers run aanaalum budget and rate limits correct-ah sync aaguma?	Distributed Mesh/Gossip consensus protocol (port 7946) synchronizing state across multi-region nodes.
Adaptive Routing	Oru AI vendor slow aana automatic-ah fastest backup vendor-ku reroute aaguma?	Real-time Multi-Armed Bandit load balancer calculating live health scores (0-100) from p95 latency.